

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech (EE) Examination

March 2018

EE 219 ANALOG ELECTRONICS**Date: 30/03/2018, Friday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below.****[07]**

- a. If the value of α is 0.9, the value of β is _____.
(i) 9 (ii) 0.9 (iii) 90 (iv) 900
- b. Operating point represents _____.
(i) Values of I_C and V_{CE} when signal is applied (ii) The magnitude of signal
(iii) Zero signal values of I_C and V_{CE} (iv) None of the above.
- c. A PN junction act as a _____.
(i) Controlled Switch (ii) Bidirectional Switch
(iii) Unidirectional Switch (iv) None of the above
- d. The value of β for a transistor is generally _____.
(i) 1 (ii) Less than 1
(iii) Between 20 and 500 (iv) Above 500
- e. The gate of a JFET is _____ biased.
(i) Reverse (ii) Forward
(iii) Reverse as well as forward (iv) None of the above
- f. A MOSFET has _____ terminals.
(i) Two (ii) five (iii) four (iv) three
- g. The best method of bias is _____.
(i) Base Resistor method (ii) Feedback resistor method
(iii) Voltage divider method (iv) None of the above.

Q – 2.a Explain the limitations in the operating conditions of PN junction.**[04]****Q – 2.b Answer any two questions.****[10]**

- (i) Define the following terms.
(1) Breakdown voltage (2) Knee voltage (3) Maximum forward current (4) Peak Inverse voltage (5) Depletion layer.
- (ii) Draw and explain the V-I characteristics of a PN junction.
- (iii) Give the difference between JFET and BJT.

Q - 3 Answer any two.**[14]**

- a. Draw pin diagram and functional block diagram of 555 timer IC. Explain function of each pin.
- b. Define stabilization and stability factor. Explain transistor as an amplifier.
- c. In potential divider biasing method of silicon transistor, it is desired to set $R_1 = 10\text{ K}\Omega$, $R_2 = 5\text{ K}\Omega$, $R_C = 1\text{ K}\Omega$, $R_E = 2\text{ K}\Omega$, $V_{CC} = 15\text{V}$. Draw the d.c. load line and determine the operating point. Take $V_{BE} = 0.7\text{V}$.

SECTION – II

- Q - 4 Answer the question below. [07]**
- State the applications of OP-AMP.
 - What are the Four types of differential amplifiers?
 - The output of differentiator for a square input is _____ wave.
(i) Square (ii) Sine (iii) Triangular (iv) Spikes
 - Give the name of different types of filter.
 - Frequency response of differentiator is same as that of _____ filter.
 - For comparator application, the OPAMP is used in the _____ mode.
(i) Open loop (ii) Closed loop
(iii) Both (i) and (ii) (iv) None of the above
 - List four different types of Regulators.
- Q – 5.a Draw the block diagram of an OP-AMP and explain the purpose of using each block. [05]**
- OR**
- Q – 5.a With the help of a neat circuit diagram explain the operation of an adjustable voltage regulator using IC LM 317. [05]**
- Q – 5.b Explain OP-AMP is used as a summing amplifier. [04]**
- Q – 5.c What are the characteristics of an ideal op-amp? [05]**
- OR**
- Q – 5.c For a first order high pass filter, Calculate the value of R if $C = 0.0047 \mu\text{F}$ and $f_c = 10\text{KHZ}$. [05]**
- Q – 6.a Draw and explain the circuit of Integrator using Op-Amp. [05]**
- Q – 6.b Define the following parameter of an Op-amp. [04]**
- Common mode rejection ratio(CMRR)
 - Power supply rejection ratio(PSRR)
 - Slew rate(SR)
 - Input offset voltage
- Q – 6.c Explain the operation of a zero crossing detector with diagram. [03]**
- Q --6.d Draw neat circuit diagram of dual input balanced output differential amplifier. [02]**
- OR**
- Q – 6.a Explain with the help of circuit diagram, the operation of first order low-pass filter. [07]**
- Q – 6.b Explain circuit of positive clippers with input and output waveform [05]**
- Q – 6.c Compare active and passive filters. [02]**
